

Mathematics for Computing

Module 3: Descriptive Statistics

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Module 3: Descriptive Statistics

Topics Covered

1. Introduction
2. Histogram
3. Sample Mean
4. Variance and Standard Deviation
5. Order Statistics
6. Sample Covariance
7. Covariance Matrix
8. Correlation
9. Frequentist Statistics
10. Sampling
11. Mean Squared Error
12. Consistency and Sampling Error
13. Confidence Intervals
14. Parametric Model Estimation
15. Non-Parametric Model Estimation
16. AI and ML Applications

What is Statistics?

INTRODUCTION

Definition:

Statistics is the science of collecting, organizing, summarizing, analyzing and interpreting data.

Statistics helps us answer questions such as:

- ▶ What is the typical value of the data?
- ▶ How much does the data vary?
- ▶ Are two variables related?
- ▶ How reliable is an estimate?
- ▶ How uncertain is a result?

AI/ML Example:

Statistics is used to understand datasets before training a machine learning model.

INTRODUCTION

Descriptive statistics summarizes the data that has already been collected.
Important descriptive measures include:

- ▶ Histogram
- ▶ Mean
- ▶ Variance
- ▶ Standard deviation
- ▶ Order statistics
- ▶ Covariance
- ▶ Correlation

Raw Data → Useful Summary

HISTOGRAM

A histogram is a graphical representation of the frequency distribution of numerical data.

The data range is divided into intervals called **bins**.

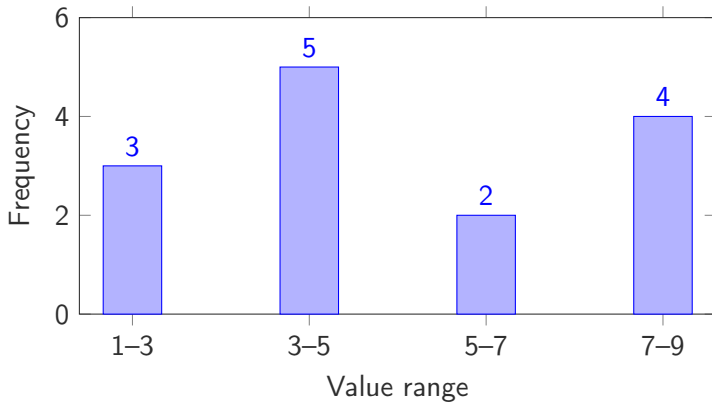
Frequency = Number of observations in a bin

Example:

$$X = \{1, 2, 2, 3, 3, 3, 4, 4, 5, 6, 7, 7, 8, 9\}$$

Histogram Example

HISTOGRAM



HISTOGRAM

- ▶ Data distribution
- ▶ Central tendency
- ▶ Spread
- ▶ Skewness
- ▶ Possible outliers

Example:

For a machine learning feature such as:

$$X = \text{Patient Age}$$

a histogram shows how ages are distributed in the dataset.

Measures of Central Tendency: Median

Definition

The median is the middle value of a dataset after sorting it in ascending order. It divides the data into two equal halves (50% below, 50% above). It is robust to outliers.

Equation:

- ▶ For odd n : $\text{Median} = x_{(\frac{n+1}{2})}$
- ▶ For even n : $\text{Median} = \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2}$

Relation with Order Statistics: $\text{Median} = X_{(k)}$ where k is middle order statistic.

Measures of Central Tendency: Mode

Definition

The mode is the value that occurs most frequently in a dataset. A dataset can have no mode, one mode (unimodal), or multiple modes (bimodal/multimodal).

Equation (for grouped data):

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where L = lower limit of modal class, f_1 = frequency of modal class, f_0 = frequency of class before modal class, f_2 = frequency after modal class, h = class size.

Measures of Dispersion: Range

Definition

Range is the simplest measure of dispersion. It is the difference between the maximum and minimum values in the dataset. Highly sensitive to outliers.

Equation:

$$\text{Range} = X_{\max} - X_{\min} = X_{(n)} - X_{(1)}$$

where $X_{(n)}$ and $X_{(1)}$ are max and min order statistics.

Measures of Dispersion: Interquartile Range (IQR)

Definition

IQR measures the spread of the middle 50% of data. It is the difference between the third quartile (Q_3 , 75th percentile) and first quartile (Q_1 , 25th percentile). It is robust to outliers.

Equation:

$$IQR = Q_3 - Q_1$$

Quartile Position: Q_1 = Median of lower half, Q_3 = Median of upper half

Box Plot

Definition

Box Plot is a graphical summary based on the five-number summary: Minimum, Q_1 , Median (Q_2), Q_3 , Maximum. It visualizes dispersion, skewness, and outliers.

Construction:

- ▶ Box is drawn from Q_1 to Q_3 , with a line at Median
- ▶ Whiskers: Extend to $Q_1 - 1.5 \times IQR$ (Lower) and $Q_3 + 1.5 \times IQR$ (Upper)
- ▶ Points outside whiskers are marked as outliers

Definition

An outlier is an observation that lies an abnormal distance from other values. It can arise due to variability or experimental error.

Detection Equations:

1. **IQR Rule:** Outlier if $x < Q_1 - 1.5 \times IQR$ or $x > Q_3 + 1.5 \times IQR$
2. **Z-score Rule:** $Z_i = \frac{x_i - \bar{x}}{s}$, if $|Z_i| > 3$, it is an outlier.

Skewness

Definition

Skewness is a measure of asymmetry of the probability distribution about its mean.

Equation (Moment Coefficient):

$$\gamma_1 = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3}$$

Sample Skewness:

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{s^3}$$

Interpretation: $\gamma_1 = 0$ Symmetric, $\gamma_1 > 0$ Right (Positive) Skewed, $\gamma_1 < 0$ Left (Negative) Skewed.

Kurtosis

Definition

Kurtosis measures the "tailedness" of a distribution relative to a normal distribution. It describes the sharpness of peak and weight of tails.

Equation (Excess Kurtosis):

$$\text{Kurt}_{\text{excess}} = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3 = \frac{\mu_4}{\sigma^4} - 3$$

Sample Excess Kurtosis:

$$g_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{s^4} - 3$$

Interpretation: $g_2 = 0$ Mesokurtic (Normal), $g_2 > 0$ Leptokurtic (Sharp peak, heavy tails), $g_2 < 0$ Platykurtic (Flat peak, light tails).

Sample Mean

SAMPLE MEAN

Let the observations be:

$$x_1, x_2, \dots, x_n$$

The sample mean is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

It represents the arithmetic average of the observations.

Sample Mean – Example

SAMPLE MEAN

Consider:

$$X = \{2, 4, 4, 5, 6\}$$

$$\bar{x} = \frac{2 + 4 + 4 + 5 + 6}{5} = \frac{21}{5}$$

$$\boxed{\bar{x} = 4.2}$$

SAMPLE MEAN

Suppose a regression model produces:

$$E = \{2, 4, 3, 5, 1\}$$

The average error is:

$$\bar{E} = \frac{2 + 4 + 3 + 5 + 1}{5}$$

$$\boxed{\bar{E} = 3}$$

VARIANCE AND STANDARD DEVIATION

Variance measures the spread of observations around the sample mean.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Large variance $\Rightarrow$ large spread.

Small variance $\Rightarrow$ small spread.

Variance – Example

VARIANCE AND STANDARD DEVIATION

$$X = \{2, 4, 4, 5, 6\}$$

$$\bar{x} = 4.2$$

Squared deviations:

$$4.84, \quad 0.04, \quad 0.04, \quad 0.64, \quad 3.24$$

$$s^2 = \frac{4.84 + 0.04 + 0.04 + 0.64 + 3.24}{4}$$

$$s^2 = 2.2$$

VARIANCE AND STANDARD DEVIATION

$$s = \sqrt{s^2}$$

For the previous example:

$$s = \sqrt{2.2}$$

$$s \approx 1.483$$

Standard deviation has the same unit as the original data.

ORDER STATISTICS

Arrange the observations in increasing order:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

Example:

$$X = \{8, 3, 10, 2, 7\}$$

$$\boxed{\{2, 3, 7, 8, 10\}}$$

ORDER STATISTICS

For odd n :

$$\text{Median} = x_{(\frac{n+1}{2})}$$

For:

$$X = \{2, 3, 7, 8, 10\}$$

$$\text{Median} = 7$$

ORDER STATISTICS

Quartiles divide ordered data into four parts.

Q_1 = First quartile

Q_2 = Median

Q_3 = Third quartile

$$IQR = Q_3 - Q_1$$

ORDER STATISTICS

Percentiles divide data into 100 parts.

$$P_{25} = Q_1$$

$$P_{50} = Q_2$$

$$P_{75} = Q_3$$

AI/ML Example:

An error above the 95th percentile is unusually large relative to most observed errors.

SAMPLE COVARIANCE

Covariance measures how two variables vary together.

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- ▶ Positive covariance: increase together.
- ▶ Negative covariance: move in opposite directions.
- ▶ Near zero: weak linear co-variation.

SAMPLE COVARIANCE

$$X = \{1, 2, 3, 4\}$$

$$Y = \{2, 4, 6, 8\}$$

As study hours increase, exam scores also increase.

$$\text{Cov}(X, Y) > 0$$

SAMPLE COVARIANCE

$$\bar{x} = 2.5$$

$$\bar{y} = 5$$

$$s_{XY} = \frac{(-1.5)(-3) + (-0.5)(-1) + (0.5)(1) + (1.5)(3)}{3}$$

$$s_{XY} \approx 3.33$$

COVARIANCE MATRIX

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1d} \\ s_{21} & s_{22} & \cdots & s_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ s_{d1} & s_{d2} & \cdots & s_{dd} \end{bmatrix}$$

where:

$$s_{ij} = \text{Cov}(X_i, X_j)$$

COVARIANCE MATRIX

$$S = \begin{bmatrix} s_X^2 & s_{XY} \\ s_{XY} & s_Y^2 \end{bmatrix}$$

Diagonal elements:

Variances

Off-diagonal elements:

Covariances

COVARIANCE MATRIX

Applications include:

- ▶ Principal Component Analysis (PCA)
- ▶ Feature analysis
- ▶ Dimensionality reduction
- ▶ Gaussian models
- ▶ Computer vision

CORRELATION

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}$$

$$-1 \leq r_{XY} \leq 1$$

$r = 1 \Rightarrow$ perfect positive relationship

$r = -1 \Rightarrow$ perfect negative relationship

$r = 0 \Rightarrow$ no linear correlation

CORRELATION

$$R = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

$$\boxed{r_{ii} = 1}$$

Every variable is perfectly correlated with itself.

CORRELATION

Example:

House Area $\longleftrightarrow$ House Price

A high positive correlation suggests a strong positive linear relationship.

Important

Correlation does not imply causation.

FREQUENTIST STATISTICS

- ▶ Parameters are fixed but unknown.
- ▶ Observed data are treated as random.
- ▶ Probability describes repeated sampling behaviour.

$$\hat{\theta} = f(X_1, \dots, X_n)$$

Frequentist Example

FREQUENTIST STATISTICS

Suppose:

$$\mu = ?$$

A sample is:

$$X_1, X_2, \dots, X_n$$

We calculate:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$\bar{X}$ estimates μ

SAMPLING

Sampling is the process of selecting observations from a population.

- ▶ Large population
- ▶ Lower cost
- ▶ Less time
- ▶ Representative data

Population $\supset$ Sample

SAMPLING

- ▶ Dataset creation
- ▶ Train/test splitting
- ▶ Cross-validation
- ▶ Bootstrapping
- ▶ Large-scale data analysis
- ▶ Balanced datasets

MEAN SQUARED ERROR

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where:

y_i = actual value

$\hat{y}_i$ = predicted value

MEAN SQUARED ERROR

$$y = \{3, 5, 7\}$$

$$\hat{y} = \{2, 6, 8\}$$

$$MSE = \frac{(3 - 2)^2 + (5 - 6)^2 + (7 - 8)^2}{3}$$

$$MSE = 1$$

MEAN SQUARED ERROR

MSE is widely used in regression.

$$\text{Actual Price} = 500000$$

$$\text{Predicted Price} = 480000$$

$$e = 500000 - 480000$$

Lower MSE generally means smaller prediction errors

CONSISTENCY AND SAMPLING ERROR

An estimator is consistent if it approaches the true parameter as sample size increases.

$$\hat{\theta}_n \xrightarrow{P} \theta$$

$$n \rightarrow \infty$$

More data $\Rightarrow$ more stable estimation

CONSISTENCY AND SAMPLING ERROR

$$\text{Sampling Error} = \bar{X} - \mu$$

Example:

$$\mu = 50$$

$$\bar{x} = 53$$

$$\text{Sampling Error} = 3$$

CONFIDENCE INTERVALS

For known population standard deviation:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where:

- ▶ $\bar{x}$ = sample mean
- ▶ σ = population standard deviation
- ▶ n = sample size
- ▶ $z_{\alpha/2}$ = critical value

95% Confidence Interval – Example

CONFIDENCE INTERVALS

$$\bar{x} = 50, \quad \sigma = 10, \quad n = 100$$

$$z_{0.025} = 1.96$$

$$CI = 50 \pm 1.96 \frac{10}{\sqrt{100}}$$

$$CI = (48.04, 51.96)$$

CONFIDENCE INTERVALS

Suppose a classifier has:

$$\text{Accuracy} = 92\%$$

A confidence interval provides information about uncertainty in the accuracy estimate.

Application:

Comparing the performance of two ML models.

PARAMETRIC MODEL ESTIMATION

A parametric model uses a finite number of parameters.

Example:

$$X \sim N(\mu, \sigma^2)$$

$$\theta = (\mu, \sigma^2)$$

PARAMETRIC MODEL ESTIMATION

Examples:

- ▶ Linear Regression
- ▶ Logistic Regression
- ▶ Gaussian Naive Bayes
- ▶ Gaussian Mixture Models

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$\beta_0, \beta_1$$

are estimated from training data.

NON-PARAMETRIC MODEL ESTIMATION

Non-parametric methods make fewer assumptions about the underlying distribution.

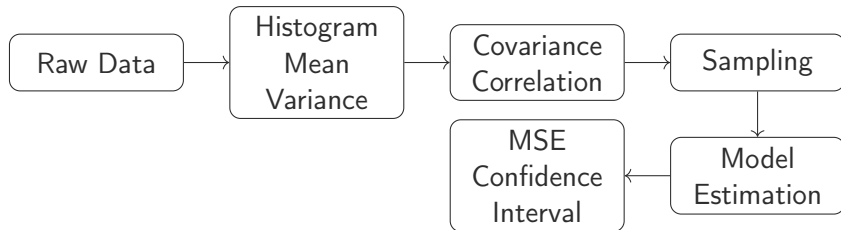
Examples:

- ▶ Histogram
- ▶ Kernel Density Estimation
- ▶ K-Nearest Neighbours
- ▶ Decision Trees

NON-PARAMETRIC MODEL ESTIMATION

	Parametric	Non-Parametric
Distribution assumption	Usually yes	Usually fewer
Flexibility	Lower	Higher
Parameters	Fixed number	May grow with data
Data requirement	Often lower	Often higher
Example	Linear Regression	KNN

AI AND ML APPLICATIONS



Example: House Price Prediction

Features:

$$X_1 = \text{Area}$$

$$X_2 = \text{Number of Rooms}$$

$$X_3 = \text{Age}$$

Target:

$$Y = \text{House Price}$$

Statistics helps us:

- ▶ Study feature distributions.
- ▶ Calculate mean and variance.
- ▶ Analyse covariance and correlation.
- ▶ Sample data.
- ▶ Estimate model parameters.
- ▶ Evaluate predictions using MSE.

AI AND ML APPLICATIONS

Features:

Age, Blood Pressure, Glucose, BMI

Statistics can be used to:

- ▶ Understand feature distributions.
- ▶ Detect outliers.
- ▶ Study feature relationships.
- ▶ Estimate parameters.
- ▶ Evaluate model performance.
- ▶ Quantify uncertainty.

Mathematics for Computing

Module 3: Descriptive Statistics

Thank You